

MARIEHAMN, Finland

WMO: 029700

Lat: 60.125N

Lon: 19.903E

Elev: 6

StdP: 101.25

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.8	-13.5	-18.9	0.7	-16.7	-15.4	1.0	-12.7	11.4	0.5	10.4	0.5	1.6	350	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	24.8	18.6	23.0	17.8	21.2	16.8	20.1	23.3	18.7	21.8	17.7	20.3	3.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.7	22.2	17.5	12.6	20.3	16.5	11.8	19.3	57.4	23.5	53.1	22.1	49.9	20.4	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 10.0	8.9	7.9		-20.0	26.9	4.6	2.2	-23.3	28.5	-25.9	29.7	-28.5	31.0	-31.9		
(5)				-20.2	20.9	4.6	1.5	-23.5	22.0	-26.2	22.8	-28.7	23.7	-32.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-1.9	-3.0	-0.7	3.7	8.7	13.0	16.5	15.6	11.8	6.5	3.2	0.3
	DBStd	7.54	4.29	5.28	4.29	2.71	3.54	2.76	2.70	2.71	2.83	3.39	3.51	4.28
	HDD10.0	1963	368	364	331	190	68	6	0	1	16	115	203	302
	HDD18.3	4447	627	598	589	439	299	161	69	90	196	366	453	560
	CDD10.0	578	0	0	0	1	27	96	202	175	70	7	0	0
	CDD18.3	21	0	0	0	0	0	1	13	6	0	0	0	0
	CDH23.3	124	0	0	0	0	2	9	88	24	0	0	0	0
	CDH26.7	12	0	0	0	0	0	0	11	1	0	0	0	0
(14) Wind	WSAvg	3.6	4.0	3.6	3.6	3.5	3.4	3.4	3.3	3.3	3.4	3.6	3.8	4.0
(15) Precipitation	PrecAvg	530	37	25	26	28	27	34	53	69	65	56	63	46
	PrecMax	783	90	85	56	59	61	90	140	125	172	151	125	92
	PrecMin	372	9	4	0	1	6	3	17	12	17	22	19	8
	PrecStd	100	21	18	12	14	18	20	33	30	33	33	25	24
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	6.8	10.0	16.1	22.2	24.7	28.1	25.9	20.2	15.0	10.5	8.7
		MCWB	4.8	3.9	6.1	9.3	15.1	16.1	19.9	19.7	16.2	12.4	9.4	7.2
	2%	DB	5.0	5.0	7.8	13.0	19.8	22.0	25.7	23.5	18.5	13.2	9.2	7.2
		MCWB	4.3	3.0	4.6	7.9	13.7	15.7	19.1	18.8	15.4	11.6	8.6	6.0
	5%	DB	4.1	3.7	6.1	10.9	17.8	20.1	23.9	22.0	17.3	12.2	8.4	6.1
		MCWB	3.5	2.8	3.6	6.8	12.3	15.0	18.5	17.9	14.8	10.8	7.8	5.2
	10%	DB	3.2	2.9	4.3	9.1	15.3	18.5	22.1	20.6	16.2	11.6	7.8	5.2
		MCWB	2.5	2.1	2.7	5.8	11.2	14.0	17.6	17.0	14.1	10.3	7.2	4.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	4.5	6.5	10.1	16.6	18.7	21.8	21.8	17.2	13.2	9.9	7.6
		MCDB	5.5	5.7	9.3	14.9	20.7	22.6	25.2	24.8	19.0	14.2	10.2	8.2
	2%	WB	4.3	3.5	5.0	8.5	14.8	16.7	20.5	19.9	16.0	12.2	8.7	6.3
		MCDB	4.7	4.4	7.1	12.0	18.6	19.8	23.9	22.5	17.7	13.1	9.0	6.8
	5%	WB	3.5	2.7	3.9	7.2	13.0	15.6	19.3	18.5	15.2	11.4	7.9	5.2
		MCDB	3.9	3.4	5.3	10.2	16.4	18.8	22.8	20.9	16.8	12.3	8.3	5.8
	10%	WB	2.6	2.1	3.0	6.0	11.3	14.8	18.2	17.5	14.7	10.6	7.0	4.4
		MCDB	3.1	2.8	4.2	8.4	15.0	17.9	21.5	19.7	16.2	11.5	7.4	5.1
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.4	6.5	7.7	8.7	8.1	8.0	7.8	6.9	5.7	4.0	4.3
		MCDBR	4.0	5.1	8.4	11.5	12.8	11.5	11.5	10.0	8.3	6.2	4.1	4.2
	5% WB	MCWBR	3.8	4.3	5.8	7.3	7.4	6.5	6.0	5.9	5.9	5.0	4.0	3.9
		MCWBR	3.7	4.2	7.2	10.7	11.3	9.4	9.4	7.9	7.1	5.3	3.8	4.0
(40) Clear-Sky Solar Irradiance	taub		0.276	0.313	0.332	0.370	0.368	0.359	0.387	0.377	0.347	0.333	0.302	0.390
		taud	2.186	2.258	2.320	2.337	2.421	2.489	2.432	2.447	2.516	2.418	2.229	2.661
	Ebn at Noon		491	680	795	827	860	874	835	815	783	659	449	347
		Edh at Noon	41	68	88	104	102	98	101	93	73	59	40	29
(44) All-Sky Solar Radiation	RadAvg		0.24	0.81	2.34	4.01	5.46	6.06	5.65	4.40	2.73	1.16	0.33	0.16
	RadStd		0.03	0.15	0.26	0.37	0.45	0.39	0.52	0.38	0.29	0.18	0.07	0.02

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	-107	N/A	N/A	N/A

Nomenclature: See separate page